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Prepared by Barbara A. Lafford

On the Teaching and Acquisition of Pronunciation within a Communicative Approach

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Abstract: In the United States, the Communicative Approach has been the focus of much intellectual debate resulting in numerous studies examining the acquisition of the four language skills. Although the acquisition of certain morphological structures and discourse strategies have received attention, studies on the acquisition of target language pronunciation have lagged behind. Recent research examining phonological instruction indicates that improvement in pronunciation for adult foreign language learners is possible by employing a multimodal methodology designed to account for individual learning style variation. An extension of this research examines experimental subjects' overall improvement in pronunciation accuracy, pinpoints specific areas where pronunciation instruction appears to be most beneficial (e.g., discrete-word repetition, sentence repetition, discrete-word reading, and free speech); and determines natural phoneme classes and specific allophones that improved as a result of phonological instruction. The findings have implications for current communicative approaches.

Key Words: acquisition, allophone, articulation, assimilation, communicative, phonology, pronunciation

1.0 Introduction

The prevailing model for foreign language instruction in the United States is the "Communicative Approach." Its aim is to create a "shift in emphasis from language knowledge to language use" (Tschirner 1996, 1) in the classroom while fostering "a broader ability to use language appropriately in natural situations" (Whitley 1993, 138). Principle avenues of investigation have involved ten major, but not all-inclusive areas of controversy surrounding this model: communicative competence (Krashen and Terrell 1983; Dulay, Burt and Krashen 1982); vocabulary building (Diller 1978; Higgs 1984; Krashen and Terrell 1983); the Monitor Model and its five sub-hypotheses (Krashen 1985; Gregg 1984); child-adult differences (Krashen, Scarcella, and Long 1982); the use of authentic materials (McGraw-Hill Teleconference 1995); communicatively-oriented syllabi (Johnson 1982); communicative testing (Canale and Swain 1979); explicit grammar instruction (Terrell 1991) and the effect of teacher/self-error correction (Frantzen and Rissel 1987; Dulay, Burt and Krashen 1982; Chaudron 1987).

While it may appear that researchers

have examined almost every facet of language acquisition in relation to the Communicative Approach, the acquisition of pronunciation has fallen to the wayside and has suffered from serious neglect in the communicative classroom. Terrell (1989, 197) notes that proponents of the Communicative Approach simply "have not known what to do with pronunciation" (1989, 197). He suggests this is due to the fact that "neither the Europeans nor the North Americans have devoted much time to the study of acquisition of sound systems." Pennington and Richards (1986, 207) note that "pronunciation, traditionally viewed as a component of linguistic rather than communicative competence or as an aspect of accuracy rather than conversational fluency, has come to be regarded as of limited importance in a communicatively oriented curriculum." In addition, Elliott (1995, 531) maintains that "teachers tend to view pronunciation as the least useful of the basic language skills and therefore they generally sacrifice teaching pronunciation in order to spend valuable class time on other areas of the language." Elliott (1995, 531) attributes this to an overall belief that pronunciation, particularly for adult learners is "more dif-

ficult to improve than other target language skills."

In spite of the scant emphasis placed on pronunciation in the communicative classroom, research examining the acquisition of target language pronunciation still managed to emerge. The most frequently cited articles fall into three categories: 1) Summative: studies that evaluate basic trends and summarize research findings (Pennington and Richards 1986; Major 1987b); 2) Theoretical: studies on variables (e.g., age, hemisphericity, universal phonology, L1 phonological transfer, task formality, Eckman's Markedness Differential Hypothesis, Major's Ontogeny Model) related to phonological acquisition and/or factors that lead to foreign accent (Snow and Hoefnagel-Hohle 1977; Scovel 1969; Eckman 1977; Major 1987a; Zampini 1994; Elliott 1995a); and 3) Experimental: empirical studies that examine the effect of phonological instruction for adult learners (e.g., Murakawa 1981; Elliott 1995b; Neufeld 1977, 1978; Neufeld and Schneiderman 1980; Suter 1976).

Studies from the latter category have yielded seemingly contradictory results indicating that phonological instruction of a second/foreign language for the adult learner can: (a) have *no* apparent relationship to pronunciation ability (Suter 1976); (b) have *beneficial* effects on pronunciation accuracy (Murakawa 1981; Neufeld and Schneiderman 1980; Elliott 1995b); (c) have possible *negative* effects on pronunciation ability leading to the overgeneralization¹ of sounds (Elliott 1995b); and (d) when not taught, have little effect or a slight non-significant negative effect on pronunciation (Elliott 1995b). Although the lack of phonological instruction in the communicative classroom does not appear to affect overall pronunciation negatively, research suggests that increasing input in the absence of phonological instruction can improve pronunciation for adult learners. For example, McCandless and Winitz (1986, 361) found that "extensive auditory input in the beginning stages of second language learning results in improved pronunciation rela-

tive to traditional procedures of language instruction." They surmise that given sufficient auditory exposure before communicating increases the likelihood of achieving native-like pronunciation. Similar findings showing positive effects of listening on pronunciation were reported by Neufeld (1978) and by Burling et al. (1981). It is logical to assume, as Terrell notes, that "the ability to produce L2 sounds and use them according to their correct patterning in the TL [target language] would automatically develop by listening to and understanding input" (1989, 207). Consequently, one would predict that the Communicative Approach with its heavy emphasis on input, should indirectly improve pronunciation through target language exposure. This may partially explain Elliott's (1995, 536) findings that not teaching pronunciation in a communicative based classroom, does not result in significantly worse pronunciation. However, counter to McCandless and Winitz (1986), Elliott (1995b) found that pronunciation when not formally taught, remained relatively stable in spite of the high degree of input the learners processed.

Another area of interest to researchers studying L2 pronunciation deals with transfer and the effect of orthography on pronunciation. Zampini (1994) found that native language transfer hampers English speakers' acquisition of the Spanish voiced spirants [b d g]. She notes that this is especially true for the Spanish allophone [d] given its phonemic status in English. In addition, Zampini found that the grapheme *v* interferes with the acquisition of the Spanish [b] and [b̞] allophones leading to less accurate pronunciation in formal reading tasks. In efforts to combat the adverse effects of transfer and spelling, Zampini recommends oral exercises that focus on allophonic and phonemic differences between Spanish and English and on differences relating to orthography as well.

Elliott (1995b) made a similar proposal to Zampini's with respect to teaching pronunciation in a communicatively oriented classroom. He shows that improvement in

Spanish pronunciation for adult learners is possible by employing a multimodal methodology that accounts for individual learning style variation. The methodology aims to promote a metalinguistic awareness based on interlingual allophonic and phonemic similarities and differences as well as an awareness of the grapheme-phoneme relationship. The present study extends this research by: a) comparing experimental and control group subjects' overall improvement in pronunciation; b) pinpointing specific areas where pronunciation instruction appears to be most beneficial (e.g., discrete-word/sentence repetition, discrete-word reading, and free speech); and c) determining which natural phoneme classes and specific allophones improved as a result of pronunciation instruction.

The findings from this research have certain implications from which current communicative approaches can benefit. First, empirical data will show that adult learners of Spanish benefit from phonological instruction when they complete exercises that focus their attention on the target language sound system. This appears to contrast with Terrell (1989, 208), who states that "learners will acquire a sound system better with meaningful input than with artificial exercises designed to focus on certain sounds or on sound patterns." Second, by pinpointing areas where students improve most, teachers can tailor their methodologies, honing in on specific areas where phonological instruction is most beneficial. Third, by examining improvement of individual sound classes and allophones it may be possible to determine which sounds are more easily acquired and how this acquisition relates to allophonic or phonemic differences between Spanish and English. This in turn, will lay down the cornerstone in formulating a theory of phonological acquisition with the eventual promise of identifying natural orders of sound acquisition. Terrell (1989, 200) notes that research investigating interlanguage phonology and the effect of explicit instruction on pronunciation will enable us to "reevaluate our aspirations for phonetic accuracy"

for the adult learner and will help to determine appropriate procedures for teaching pronunciation in the communicative classroom.

2.0 Experimental design and methodology:

The present study addresses the following questions:

1. Does formal phonological instruction relate to improved pronunciation of the 19 sounds for the experimental group subjects?
2. Is formal instruction more beneficial in improving Spanish pronunciation for: a) word repetition, b) sentence repetition, c) word reading, or d) spontaneous production of the target language?
3. Which natural phoneme classes and specific allophones improved significantly as a result of phonological instruction? Are these sounds similar or different in comparison to their English counterparts? For example, do subjects show more improvement when learning to pronounce the Spanish trilled *r* than they do when learning other sounds that are more similar to English (e.g., [g k b])? A preliminary examination of the graphemes *b* and *v* will provide indications as to the effect of spelling on pronunciation with regard to the Spanish allophones [b] and [ɸ] and the English phoneme /v/.

2.1 Subjects and sample parameters

The data were collected during the fall semester of the 91–92 academic year at Indiana University, Bloomington. Sixty-six undergraduate students enrolled in three sections of an intermediate Spanish course participated as small intact comparison groups.

Two classes made up the experimental group and one class served as the control section. The experimental group had forty-three subjects: twenty males and twenty-three females. The control group consisted of twenty-three subjects: twelve males and eleven females. The experimental group curriculum differed only in terms of the pronunciation instruction that was pro-

vided. All other aspects of the course were similar to other third semester Spanish courses taught at this university. Class sessions were held three days a week for a period of fifty minutes each throughout the semester.

2.2 The pronunciation pretest and posttest

The data for both the experimental and control groups came from a pronunciation test (Elliott 1995a) administered during the second week of classes and again at the end of the semester. The test consisted of four sections,² each believed to measure specific skills relating to one's ability to pronounce Spanish. Section 1³ measured accuracy in mimicking pronunciation⁴ at a *discrete* word level; Section 2 measured accuracy in mimicking pronunciation at a *sentence* level; Section 3 measured accuracy of pronunciation of isolated written words, and Section 4 was a free elicitation exercise requiring subjects to describe one of two pictures in Spanish for approximately a minute and a half. Figure 1 contains the descriptive statistics for both the pretest and posttest.

Allophones were selected from a survey of several Spanish instructors, professors, and graduate students who listed Spanish sounds they felt contributed most to a foreign accent. Foreign accent, in this instance, consists of any oral linguistic output that a native Spanish speaker would deem as *sounding* unnatural or nonnative. Survey respondents consisted of native or near-native speakers of Spanish. Respondents listed the following 19 sounds as contributing most to a foreign accent for American English speakers learning Spanish: [rr, r, **ɾ**, **ɰ**, **g**, b (represented graphemically by b and v), p, t, k, d, ñ, z (voicing of s before a voiced consonant as in [míz-mo]), m of n phoneme (assimilation of n to a bilabial as in [um-pé-so]), g, a, e, i, o, u]. Because the instructors and professors surveyed have dealt mainly with students from the United States, the sounds chosen reflect pronunciation problems most encountered by adult native American English speakers learning Spanish.

2.3 Scoring procedures, subsection correlations, test reliability

Recordings of the pronunciation pretest and posttest were randomly ordered so the judges did not know which test they were grading or the section it was from. Three judges listened to tapes of the subjects' pronunciation approximately 16 weeks after the end of the semester. The first judge is Chilean, working toward her Ph.D. in Spanish Linguistics at Indiana University, Bloomington. The second judge is from the United States and has a B.A. degree in Spanish from the Ohio State University. He lived seven years in Bogotá, Colombia, possesses native fluency in Spanish, and has native/near-native Spanish pronunciation. The experimenter, who was working on his doctorate in Spanish Literature and Linguistics, served as the third judge. The experimenter lived approximately a year and a half in Bogotá, Colombia and a half year in Spain and has native/near-native Spanish pronunciation.

The judges attended a practice scoring session in order to clarify the criteria for scoring the subjects' pronunciation. The judges scored the allophones strictly on the basis of authenticity: how native-like the sounds were. Similar to scoring for the Bilingual Syntax Measure (Burt, Dulay, and Hernández-Chávez 1975), the judges awarded the subjects a score ranging from one, incorrect target sound, to three, correct target sound. A score of two indicated the subject made a close approximation of the target sound (e.g., use of a uvular r as opposed to the English retroflex). For Section 4, the free elicitation portion of the test, the judges awarded overall scores ranging from 5 (native accent) to 1 (almost unintelligible).

Each allophone appeared at least once in each section of the pretest and posttest; thus providing minimally 3 obligatory contexts for the majority of sounds. Three allophones: the ñ, the m of the n phoneme (i.e., [um-pé-so] for un peso), and the s before a voiced consonant (i.e., [míz-mo] for mismo), were tested twice on both tests. Sounds (e.g.: [r, rr, **ɾ**, **ɰ**, **g**]) that survey re-

spondents listed more than other sounds were believed by this researcher to be more difficult universally for American English speakers learning Spanish and appeared two or more times in each section of the test; thus providing approximately five or six obligatory contexts per sound. There were 65 items on both the pretest and posttest.

SPSS Correlations for the four sections on the pretest ranged from .66 to .95 and for the posttest from .69 to .96. All correlations were significant at the .05 level. A split-half reliability coefficient of .88 was obtained for the pretest and .92 for the posttest (Figure 1). An independent-samples t-test indicated that subjects of the experimental and control groups were not significantly different in their pronunciation at the beginning of the semester ($t = -.06$, $p = .949$).

2.4 Interrater reliability

Split-half interrater reliability coefficients ranged from .96–.97 for the pretest and from .97–.98 for the posttest. All correlations were beyond the $p < .001$ significance level indicating that the judges' ratings of the subjects' pronunciation were very consistent.

3.0 Pronunciation instruction

Experimental group subjects received 10 to 15 minutes of formal instruction and practice of Spanish pronunciation for a total of twenty-one class periods. The instructor taught the following allophones both deductively and inductively: [a e i o u r rr b b̃ d ð g ɣ p t k m (of the n phoneme) ñ z (of the s phoneme)]. The pronunciation sessions began with a model word or phrases that contained the allophone under study. Subjects described the allophones in terms of point, place, and manner of articulation in their own words (i.e., without using linguistic terminology) or drew facial diagrams on the board while explaining where they believed their tongue would hit when producing the sound. Once the students successfully described the allophone and contrasted it with its English counterpart, the subjects then

compared their description with a transparency of a facial diagram. The pronunciation instruction was designed to appeal to individual differences in learning styles and preferences (Elliott 1995b). Subjects completed practice pronunciation exercises such as word and sentence repetition exercises, rhymes, and tongue twisters (e.g., *Erre con erre cigarro...*). Students learned information on the graphemic representation of the sound (e.g., the graphemes “b” and “v” are pronounced as “b”) and how pronunciation of specific sounds varies due to positional constraints (e.g., how pronunciation of the Spanish postdental “d” differs in initial, medial, and final position).

4.0 Research questions and findings

4.1 Overall improvement in pronunciation

The first question examines overall improvement in pronunciation for the 19 sounds. An Analysis of Covariance⁵ (ANCOVA) was performed with the pronunciation posttest as the dependent variable and the pretest as the covariate. The independent variable consisted of a dummy variable GROUP in order to test for the incremental effects of the treatment on pronunciation ability for the experimental subjects. (Control subjects were classified as 0 and experimental subjects as 1.) Pretest scores for both groups correlated significantly with the posttest scores, thus validating the use of an ANCOVA analysis. Figure 2 presents the results of the ANCOVA analysis based on adjusted means. The covariate pretest yielded an F-ratio of $F = 68.26$, $p < .0001$. Once the pretest variation in pronunciation was removed, an F-ratio of $F = 19.08$, $p < .0001$ was found for the effects of the pronunciation instruction. The results suggest that formal phonological instruction relates to and perhaps promotes more accurate Spanish pronunciation.

4.2 Improvement by task.

The next question attempts to determine improvement in pronunciation by specific tasks. In other words, do subjects in the experimental group show statistically signifi-

cant improvement in pronunciation when: a) repeating isolated words after a native-speaker model; b) when repeating sentences after a native-speaker model; c) when reading isolated words; or, d) when spontaneously communicating in Spanish?

Scores for each section of the pronunciation posttest served as the dependent variables in this analysis. The corresponding pretest scores (Sections 1–4) served as the concomitant variables. The dummy variable differentiating experimental and control group subjects was the independent variable. Figure 3 presents the results of the four ANCOVA analyses based on adjusted means. Note that similar to the previous ANCOVA analysis, the observed F-ratios for the various sections of the pronunciation pretest in relation to the corresponding posttest sections ranged from 34.23 to 89.95, indicating again that posttest variance equals initial ability (pretest variance) plus the differential effects of the instruction provided. The results in Figure 3 suggest that the experimental group subjects experienced statistically significant improvement in pronunciation for Section 1, the word repetition exercise ($F = 11.16, p < .001$); for Section 2, the sentence repetition exercise ($F = 15.11, p < .0001$); and for Section 3, the word reading exercise ($F = 17.07, p < .0001$). An F-ratio denoting improvement in pronunciation for the free elicitation portion of the test approached statistical significance ($F = 3.79, p < .056$). The results of the present analysis suggest that phonological instruction is most beneficial for subjects when reading Spanish words on a discrete-word level followed by sentence repetition and then for word repetition also on a discrete level. Once again, there was a slight tendency for improvement in pronunciation when spontaneously producing the target language; however, the results were not statistically significant.

4.3 Improvement of natural classes and individual sounds.

SPSS post-hoc Scheffé procedures tested for improvement in pronunciation of natural sound classes (e.g., stops, fricatives,

nasals, liquids, and vowels). The findings in Figure 4 indicate that the experimental subjects experienced significant improvement of the liquids ($F = 16.76, p < .0001$) and of the stop phonemes ($F = 7.93, p < .006$). Significant improvement in pronunciation did not result for their pronunciation of the fricatives ($F = 2.09, NS$) or of the nasals ($F = 3.40, NS$). The vowels, when grouped together, approached statistical significance yielding an F-ratio of $F = 3.81, p < .055$.

ANCOVA analyses served to explore the effects of the treatment on individual allophones. The DUMMY variable group was regressed onto the posttest scores for the individual allophones using the pretest scores as covariates. Note in Figure 5 that statistically significant differences denoting improvement in pronunciation were found for the experimental group's articulation of the following allophones: [rr p r d b t u k o ñ]. Teachers and researchers tend to believe that the Spanish trilled r is the most difficult sound for English speakers. Surprisingly, the Spanish trill experienced the most improvement yielding an F-Ratio of $F = 17.31, p < .0001$. Experimental subjects improved their pronunciation of the intervocalic flap ($F = 9.48, p < .003$). The stop allophones [p t k] also improved significantly indicating that experimental subjects had a greater propensity to *not* aspirate these sounds by semester's end. The back vowels [o u] improved significantly, whereas the central and front vowels [a e i] showed no significant improvement. Sounds undergoing point of articulation assimilation (e.g., *um-pé-so* for *un peso* and *míz-mo* for *mismo*) were more difficult concepts that students had not mastered by the end of the semester. All other allophones, although demonstrating a slight upward trend in improvement, did not change significantly as a result of the instruction.

In order to examine pronunciation accuracy in relation to the graphemes *b* and *v* the data were divided into three groups: 1) orthographic *b* realized as [b], 2) orthographic *b* as [β], and 3) grapheme *v* as [b] and [β]. Mean raw scores for these graphemes and their corresponding targeted

allophones were ranked from highest to lowest for experimental and control subjects on both the pronunciation pretest and posttest. The same rank orders were found for the experimental and control subjects on both the pre- and posttests. The grapheme *b* as a stop allophone ranked first. This finding suggests that the subjects experienced less difficulty when pronouncing the Spanish stop [b] when represented graphemically as *b*. There were no instances in which subjects pronounced the grapheme *b* as the English *v*. Orthographic *v* ranked second on both tests. The majority of errors resulted from pronouncing this grapheme as a voiced labiodental fricative [v]. This finding was true whether the target allophone called for a stop or a non-fricative continuant. The voiced non-fricative continuant [ɸ] in words containing the grapheme *b* consistently ranked the lowest for accuracy; however, it was the only non-fricative continuant to improve significantly for experimental subjects. This tentative ranking of difficulty suggests that mastery of the Spanish [ɸ] is more difficult than is learning to pronounce the grapheme *b* as a stop or learning to pronounce orthographic *v* as [b] or [ɸ].

5.0 Discussion

The present study reveals several significant findings regarding phonological instruction for adult learners of Spanish. In the first analysis, the results suggest that formal instruction in pronunciation was significantly related to overall improvement for the experimental group. A major implication for the Communicative Approach is that students can benefit from pronunciation instruction when they engage in exercises requiring them to focus on the target language sound system. As previously stated, researchers maintain that increasing auditory input in the early stages of language learning leads to improved pronunciation for the adult foreign language learner. In spite of the high degree of input control subjects processed, little change occurred in their pronunciation during the

semester (Elliott 1995b). On the other hand, the same curriculum with the pronunciation component added resulted in significant improvement for the experimental group.

In the second analysis, the finding that phonological instruction relates to improved pronunciation when reading isolated Spanish words provides an indication that orthography might possibly be an impediment to phonological acquisition. Empirical evidence offered by Zampini (1994, 479) suggests that spelling may “adversely affect pronunciation in L2.” The methodology used in the present study promoted an awareness of the grapheme/phoneme relationship in an attempt to minimize the influence of spelling on pronunciation. By teaching students the graphemic representation of the sound (e.g., graphemes ‘b’ and ‘v’ are pronounced as ‘b’; intervocalic ‘r’ sounds like the ‘t’ in city), and how pronunciation of specific sounds varies due to positional constraints (e.g., the *n* assimilating to a bilabial consonant or the voicing of the *s* before a voiced consonant), one can lessen the influence spelling apparently has on pronunciation.

In the present study, formal phonological instruction related to improved pronunciation for experimental subjects when imitating sounds on a discrete-word level and when repeating sentences after a native speaker model. Students experienced little difficulty when imitating the non-fricative continuants [b d g] and the flap ‘r’; however, the tendency was to use English sounds when producing more linguistically complex structures (e.g., the free elicitation exercise). This finding supports Major’s Ontogeny Model (1987) in which transfer errors are more frequent in casual speech but decrease as speech becomes more formal. Major (1987, 107) states that “speakers are able to correctly produce sounds and words in isolation, but in running speech they slip back into L1 patterns.” The free elicitation exercise is less formal than the word/sentence repetition or word reading exercises. As students focused their attention on communicating meaning as op-

posed to their pronunciation, more transfer errors emerged (e.g., retroflexion of [r] and [rr], diphthongization, vowel lengthening, the use of stops in fricative environments, etc.).

The finding that more transfer errors resulted in the free elicitation portion of the pronunciation pretest and posttest supports Tarone (1979), who states that the amount of attention learners pay to their speech influences L2 pronunciation. Tarone (1983) argues that a learner's interlanguage will evidence systematic variability on a continuum ranging from careful to vernacular speech styles. Vernacular styles produce "the most regular and systematic phonological and grammatical patterns" (Tarone 1979, 181), whereas careful styles are prone to target and native language interference. Tarone (1979, 186) argues that whenever "a speaker is systematically observed," careful speech will result. Because the subjects in the present experiment were aware they were being tape-recorded their speech was not "monitor-free" and constituted a more careful style that according to Tarone would evidence more target and native language interference—a finding that was borne out in the present research.

In the final analyses, experimental group pronunciation improved significantly for the liquid and stop phonemes. Of the liquids, both the trill and the flap improved significantly. The methodology used in this experiment was designed to capitalize on allophonic similarities and differences between the two languages. Experimental subjects learned to pronounce the intervocalic 'r' replicating the flap in words like *butter*, *city*, and *water*. Thus, the improvement of the flap 'r' is probably best explained by positive transfer. On the other hand, improvement of the voiced alveolar trill supports Hardy (1994), who found that it was easier for her subject to learn new sounds not present in his native language. Learning to use the English *v* was easier than it was to correct errored pronunciation due to positional constraints (e.g., learning to pronounce *z* in word initial position) or to differences in allophonic rules between

the two languages (e.g., learning when/when not to use fricative *d*). Sounds that differ in comparison to English due to positional constraints, such as the *n* assimilating to a bilabial or the voicing of the *s* to a voiced consonant, evidenced little to no improvement in the present study in spite of the fact that pronouncing these sounds should be easy for native English speakers. The lack of improvement of these features might be explained by Eckman's Markedness Differential Hypothesis (1977). Assimilatory processes are marked and occur more frequently in Spanish than in English in syllable and word-final position. Theoretically, that would make these Spanish sounds harder to master for English speakers. The results of the present study support this claim in part; however, more empirical research is necessary.

Markedness might also explain the acquisition of the non-fricative continuants. The intervocalic 'b' was the only fricative to improve. On a universal scale, Jakobson (1968) notes that fricatives are more marked than their stop counterparts and consequently are more difficult to acquire. Of interest to note is that some subjects in the experimental group evidenced idiosyncratic behavior by replacing the stop *b* with the fricative variant thus, resulting in overgeneralization of [b]. Others sporadically used the stop and fricative *b*'s ([b b]) in free variation without any noticeable consistency. Additionally, the finding that orthographic *v* negatively influences pronunciation further supports Zampini (1994). Overall scores for Spanish words containing the letter *v* were consistently lower in comparison to words containing the grapheme *b*. As evidenced by the rankings of the graphemes *b* and *v*, learning to pronounce the letter *v* as [b] was not as difficult as was learning to pronounce the non-fricative continuant [b].

All voiceless stops improved significantly. Experimental subjects learned to monitor their production of [p t k] without aspirating as in English. The only voiced stop to improve was the [d] allophone. Correct pronunciation of the Spanish [d] and [t]

required students to change the point of articulation from an alveolar to a postdental position. Students not only had to eliminate aspiration after [t]; they also had to change its point of articulation. Consequently, students had to learn more in order to pronounce this sound correctly. This might explain the low but statistically significant F-ratio obtained for [t], ($F = 5.89, p < .018$).

The vowels improved slightly but, as a group, did not achieve overall statistical significance ($F = 3.81, p = .055$). Recurring problems with the vowels included: diphthongization of mid vowels, vowel lengthening and centralization. Mispronunciations of this nature tend to be features that usually linger on in the learner's speech beyond the first year. The ANCOVA analyses indicated that only the back vowels, /o/ and /u/, improved significantly. These data might be suggestive of an acquisition order in which the back vowels are acquired before central or front vowels; however, stronger evidence might lie with universal phonological principles. Jakobson (1968, 56) notes that the "pair o-u cannot...precede the pair i-e" and that "there are no children who have an o-phoneme without having acquired an e-phoneme." Perhaps at this point in the students' developing phonological system, the front and central vowels had already been acquired. Thus, the degree of improvement for the vowels [a e i] was not proportionally equal to the total degree of change possible for the o-u vowels. Future research focusing on learners in the initial stages of language learning may provide a more complete picture of the acquisition of target language vowels for adult learners of Spanish.

6.0 Conclusions and implications for future research

The results of the present study imply a need for formal phonological instruction at the intermediate level. The findings suggest that teaching pronunciation to adults is beneficial in the acquisition of Spanish pronunciation as evidenced by the experimental group subjects' overall improve-

ment in pronunciation over the course of the semester. Subject pronunciation improved significantly for the word reading, the sentence repetition, and the word repetition exercise. Experimental group pronunciation of the free elicitation portion of the pronunciation test approached statistical significance. These findings justify a continued emphasis in the search for ways to improve L2 pronunciation.

Although the current study demonstrates that instruction in pronunciation can be a positive addition to a communicative approach, it did not, however, examine the effect of phonological instruction in relation to other methodologies. Future studies might benefit from examining the effect of formal instruction in pronunciation as it relates to audiolingual, cognitive-code, and proficiency based classrooms.

As pronunciation gains more attention, future research may reveal that phonological instruction results in additional benefits such as enhanced listening comprehension and communication skills. Zampini (1994) suggests that communication with native speakers may be hampered when students do not recognize allophonic and phonemic interlingual differences. She states that if an "English-speaking student pronounces the Spanish word, *todo* as [tóDo], native speakers of Spanish may in fact perceive [tóro]" (Zampini 1994, 471). Furthermore, Okamura (1995, 34) found that native speakers ranked pronunciation as a major factor for "distinguishing good from poor speakers of Japanese." By incorporating pronunciation exercises designed to focus specifically on those Spanish sounds that are most problematic for native English speakers and negatively influence L2 pronunciation, it may be possible to enhance the quality of native and non-native speaker communication. Previous research (Zampini 1994; Okamura 1995) seems to support such a claim.

Finally, teachers and researchers have long recognized the importance of affect in language acquisition. Harlow and Muyskens (1994, 146) note that students "worry about pronunciation a great deal

because they feel insecure about how they sound to other people." Additionally, Suter (1976) and Elliott (1995a, 1995b) found that students who were more concerned about sounding native-like had better pronunciation. Teaching pronunciation early on may increase student concern for developing native/native-like pronunciation, lower their affective filters, and help students to feel less anxious about speaking. With renewed confidence in the way they sound, students might be more motivated to actively seek out native speakers with whom to converse. More contact with native speakers translates to greater amounts of target language input and negotiation of meaning, which in turn, will serve to enhance our students' ultimate degree of acquisition, thus complying with the principal goal of communicatively-oriented curricula.

■ NOTES

¹It is generally accepted in the field that overgeneralization errors indicate progress in the learner's interlanguage or approximative system.

²The original pronunciation test contained 84 items (Elliott, 1995a). Sounds (e.g., ll, ei, ue, ui, qui) that were not listed by the majority of the survey respondents were excluded from the present analysis.

³The allophones Section 1 of the pronunciation test were: GANAR, BEBER, CADA, CIERRA, DESDE, PERO, MAÑANA, ROPA, UN PESO, FUERTE, JOTA, HECHO, LIBRO, CABALLO, and PEGAMOS.

⁴A native Panamanian woman recorded the oral cues for Sections 1 and 2 of the pronunciation test.

⁵Alpha levels for both the ANCOVA and SCHEFEE procedures were set at .05 and .10 respectively.

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Figure 1. Descriptive statistics of the Pronunciation Pretest and Posttest

STATISTIC				
VARIABLE	# OF ITEMS	$\bar{X}$	S	REL.
PRON. PRETEST	65	126.92	14.75	.88
PRON. POSTTEST	65	130.08	18.62	.92

N=66

Figure 2. ANCOVA Analysis with Treatment as the Main Effect and the Pronunciation Pretest as the Covariate (using adjusted mean scores)

Source of Variation	SS	df	MS	F	p
REGRESSION (PRETEST)	37646412	1	37646413	68.26	.0001
GROUP (TREATMENT)	10520775	1	10520775	19.08	.0001
WITHIN CELLS (ERROR)	34746406.87	63	551530		

Note. (N = 66) PRETEST = Pronunciation Pretest; GROUP = Dummy variable differentiating control and experimental group subjects; DEPENDENT VARIABLE = Pronunciation Posttest.

Figure 3. ANCOVA Analyses with Treatment as the Main Effect and Pronunciation

Pretest Scores by Task as the Covariates

Source of Variation	SS	df	MS	F	p
SECTION 1: WORD REPETITION					
REGRESSION (PRETEST)	659.07	1	659.07	34.23	.000
GROUP (TREATMENT)	214.86	1	214.86	11.16	.001
WITHIN CELLS (ERROR)	1212	63	19.25		
SECTION 2: SENTENCE REPETITION					
REGRESSION (PRETEST)	1847.88	1	1847.88	71.36	.000
GROUP (TREATMENT)	391.33	1	391.33	15.11	.000
WITHIN CELLS (ERROR)	1631.44	63	25.90		
SECTION 3: WORD READING					
REGRESSION (PRETEST)	3349.24	1	3349.24	89.85	.000
GROUP (TREATMENT)	636.20	1	636.20	17.07	.000
WITHIN CELLS (ERROR)	2348.27	63	37.27		
SECTION 4: FREE ELICITATION					
REGRESSION (PRETEST)	13.26	1	13.26	37.83	.000
GROUP (TREATMENT)	1.33	1	1.33	3.79	.056
WITHIN CELLS (ERROR)	22.08	63	.35		

Note. (N = 66)

Figure 4. Post-hoc Scheffé Analyses Examining Improvement in Pronunciation by Natural Classes

Source of Variation	df	SS	MS	F	p
STOPS:					
BETWEEN GROUPS	1	846.54	846.54	7.93	.006
WITHIN GROUPS	64	6832.14	106.75		
TOTAL	65	7678.68			
FRICATIVES:					
BETWEEN GROUPS	1	211.64	211.64	2.09	.15
WITHIN GROUPS	64	6464.83	101.01		

TOTAL	65	6676.48			
NASALS:					
BETWEEN GROUPS	1	391.12	391.12	3.40	.06
WITHIN GROUPS	64	7357.77	114.96		
TOTAL	65	7748.90			
LIQUIDS:					
BETWEEN GROUPS	1	3285.58	3285.58	16.76	.0001
WITHIN GROUPS	64	12542.28	195.97		
TOTAL	65	15827.86			
VOWELS:					
BETWEEN GROUPS	1	1928.08	1928.08	3.81	.055
WITHIN GROUPS	64	32320.92	505.01		
TOTAL	65	34249.00			

Figure 5. Example of ANCOVA Analyses with Treatment as the Main Effect and Pronunciation Pretest Scores for Individual Allophones as the Covariates (N = 66)

Source of Variation	SS	df	MS	F	p
STOP [d]:					
REGRESSION (PRETEST)	8.22	1	8.22	29.10	.000
GROUP (TREATMENT)	2.34	1	2.34	8.30	.005
WITHIN CELLS (ERROR)	17.80	63	.28		
LIQUID [rr]:					
REGRESSION (PRETEST)	5.61	1	5.61	78.49	.000
GROUP (TREATMENT)	1.24	1	1.24	17.31	.000
WITHIN CELLS (ERROR)	4.50	63	.07		
ALL OTHER SOUNDS TESTED	F-RATIO		p		
STOPS:					
[p]	16.58		.0001*		
[k]	5.07		.02*		
[t]	5.89		.01*		
[b]	.22		.64		
[g]	1.70		.19		
FRICATIVES:					
[d]	1.57		.21		
[g]	.67		.42		
[b]	6.30		.01*		

[z]	1.37	.24
NASALS:		
[m] of n phoneme	.48	.49
[ñ]	4.40	.04*
LIQUIDS:		
[r]	9.84	.003*
VOWELS:		
[a]	.58	.44
[e]	3.16	.08
[i]	.00	.98
[o]	4.48	.03*
[u]	5.23	.02*

NOTE: For space considerations, only F-Ratios and probability levels are listed.